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To cite this article: Jörg Reichardt and Stefan Bornholdt *J. Stat. Mech.* (2007) P06016

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Clustering of sparse data via network communities—a prototype study of a large online market

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Received 22 September 2006

Accepted 31 May 2007

Published 25 June 2007

Online at stacks.iop.org/JSTAT/2007/P06016

[doi:10.1088/1742-5468/2007/06/P06016](https://doi.org/10.1088/1742-5468/2007/06/P06016)

Abstract. Market segmentation of an online auction site (www.ebay.de) is studied using a novel clustering technique via community detection. A network of bidders connected by common interest in individual articles is constructed, whose community structure corresponds to the main user groups according to common interest. A key feature of the analysis is its independence of any kind of measure of similarity between the bidders or articles offered, or any kind of dimensionality reduction possibly biasing the analysis. Further, the method works on the sparse raw data directly, is scalable to large systems and can be used to discover both hierarchical and overlapping cluster structures. Results are compared to null models based on random networks and clusters are validated and interpreted using the taxonomic classifications of eBay categories. We find clear-cut and coherent interest profiles for the bidders in each cluster. The interest profiles of bidder groups are compared to the classification of articles actually bought by these users during the time span 6–9 months *after* the initial grouping. Their temporal stability indicates typical interest profiles in society. Our results shed some light on typical characteristics of online markets and their segmentation. They show how network theory can be applied successfully to problems of cluster analysis in economic and sociological milieu studies with large, sparse and high dimensional data.

Keywords: random graphs, networks, socio-economic networks, new applications of statistical mechanics, online dynamics

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1. Introduction

The internet has changed the way people communicate, work, and do business. One example is provided by online auction sites, the largest being eBay with its more than 150 million registered users worldwide [1]. An interesting aspect of eBay's success is its transparency. The market is fully transparent as the trading history of every user is disclosed to everyone on the internet. We here investigate a network of competitors in this market and use it to cluster users into groups with homogeneous buying interest profiles, so called market segments. In economics, market segmentation studies are often the starting point in designing targeted marketing campaigns and the quality and validity of this exploratory data analysis is crucial for their success. However, such studies often have to deal with several difficulties: first, a specific similarity measure has to be tailored and, second, if the data are very high dimensional and sparse it requires dimensionality reduction to make conventional methods applicable. This, however, may introduce bias in the analysis. Here, we present a method which circumvents these problems by working directly on the raw data and without the introduction of any similarity measure. Our results show how market participants use the online auction site and where there is economic growth potential for the host apart from increasing the number of users of the site.

Let us first recall the operating principle of an online auction in figure 1. Users may offer goods through the online platform and set a deadline when their auction will end. Articles are listed under a certain taxonomic product category by the seller and are searchable platform wide. Users with a particular demand either browse through the articles listed in an appropriate category or search for articles directly. Until the end of the auction they may bid on the article. The user with the highest bid at the end of the auction wins (so called hard-close) and buys the article. In every new auction, users may assume different new roles as sellers, bidders or buyers. The market can be represented as a graph with the users and/or articles as the nodes and the links denoting their interactions as shown in figure 1.

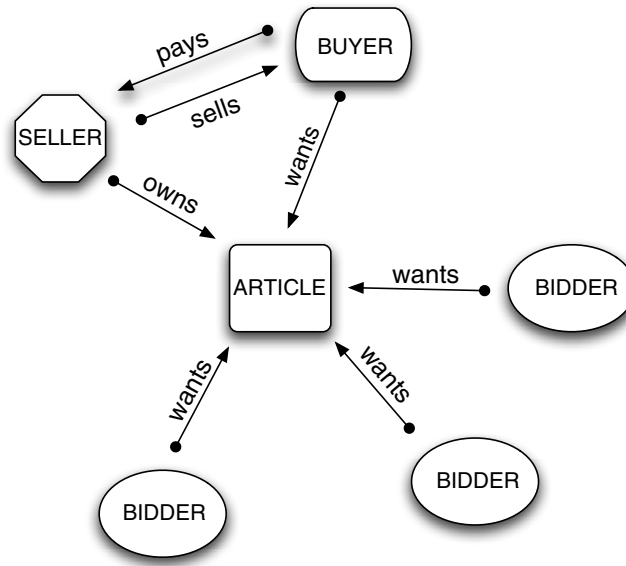


Figure 1. Structure of a single auction. Users wanting a particular article express their (common) interest by bidding. The user with the highest bid wins the auction and exchanges money and the article with the seller. EBay earns a fee with every transaction. Users of the auction site, i.e. bidders, buyers or sellers, may change their role in a different auction of another article.

A number of researchers have presented statistical studies of trading [2] and analyses of bidding strategies and auction ending rules [3, 4]. In this contribution we focus on the market segmentation of the eBay auction site. Our approach is based on the assumption that at a certain level of abstraction the population of consumers can be separated into relatively clear-cut and homogeneous sub-groups corresponding to certain customer milieus or market segments [5]. We assume that customers of the same type are described by a common pattern in their consumer interests which leads to a higher probability of bidding for the same article [6].

In particular, we perform a cluster analysis [7, 8] of the bidding behaviour of about one million users. Groups of eBay users with common interest or demand are detected using solely the information of which users competed in the same auctions. The classification is based on a very sparse and high dimensional two mode data set with only slightly more than 3 out of 1.6 million possible auctions on average for each of the 1.9 million bidders.

Classical clustering techniques generally fail for such high dimensional data due to the ‘curse of dimensionality’ [9], a problem which arises when the dimension D of the data set to be clustered increases [10]. The data points become increasingly sparse as the dimensionality increases and the relative difference between the closest and the furthest neighbour of a randomly selected point in the data set goes to zero with increasing dimension [11, 10]. In our case, the dimension of the space of articles is 1.6 million in which we find 1.9 million data points corresponding to the bidders.

Other conventional analysis techniques for two mode data such as correspondence analysis [5, 12] have to make use of a similarity measure between articles in order to reduce the dimensionality and coarse-grain the data, such as exploiting the annotation of articles

into product categories. However, this bears several pitfalls: First, the annotations are defined by the seller who lists the article such that it can be found efficiently, hence, the categorization is mainly a taxonomy. Using this to coarse-grain the data would introduce a bias in the analysis. Second, eBay categories differ largely in size when counting the number of articles in the category as well as the number of sub-categories. Correcting for this again may introduce a bias. Third, using the category taxonomy for coarse-graining induces a hierarchy in the data, as all articles below the cut in the taxonomy tree are subsumed. Fourth and most importantly, it is not clear at which level in the category tree a coarse-graining should be performed and whether this level should be the same for all branches.

Therefore, instead of using the two mode data, we project the bipartite bidder–article graph onto its one mode representation in which bidders competing for the same article are connected by links representing the different auctions in which two bidders have met. Though still sparse, i.e. with only few connections per bidder, this network exactly represents the proximity information of the original data and can be clustered directly. As will be shown, this procedure circumvents the problems of conventional analysis techniques mentioned above. Our clustering technique uses an analogy to a physical spin system, the so called Potts model [13], and is inspired by an earlier paper due to Blatt *et al* [14]. In contrast to their super paramagnetic clustering, our method does not need to keep track of a full $N \times N$ matrix of spin–spin correlations, with N being the number of nodes in the network, since this would be prohibitive given the size of our data set. Instead, we can make direct use of the sparse representation of the bidder network and we only need to store a cluster index for each node plus a histogram of cluster indices as will be outlined below. Our analysis is independent of taxonomic categories and dimensionality reduction. It allows for hierarchical and overlapping cluster structures, and we find evidence for both. The product categories are solely used to interpret the results of our study, i.e. provide interest profiles of user groups found in terms of this taxonomy.

By clustering users directly according to a common demand spectrum, we also circumvent problems of conventional basket analysis done by frequent item sets [15]–[18]. The latter asks which articles are frequently demanded by a single person. This analysis is performed for all articles averaging over the entire population of consumers and hence results in the least common denominator of articles which may then be bundled together and marketed together to the whole population of customers. The same is true for cluster analysis of eBay categories [4]. The proposed *network cluster analysis*, however, reveals information about people and their diverse and possibly very special interests.

2. Data set

A data set consisting of over 1.59 million auctions was obtained from the German eBay site www.ebay.de ending during the pre-Christmas season between 6 and 20 December 2004. Considering only articles with locations in Germany, we recorded the user-id of seller, buyer, and all bidders competing in each auction, as well as the individual bids and the product category in which the article was listed (excluding articles listed in the real estate category which was in a beta testing phase at the time). Since auctions last between 7 and 10 days depending on the choice of the seller, we thus cover a bidding period of up to 25 days. We believe the pre-Christmas time is a suitable time for analysis

Table 1. Divisions of users into their roles on eBay as observed between 6 and 20 December 2004. Numbers in millions.

Auctions observed	1.59
Users acting as buyer	0.95
Users acting as seller	0.37
Users acting as bidder	1.91
Users acting as seller and bidder	0.14
Users acting as seller and buyers	0.08

for the following reasons: First, traffic is very high. In fact, there was a broad advertising campaign in Germany advertising to shop for Christmas presents on eBay. Second, we only considered auctions and expect that users are unlikely to bid for articles for which they cannot assess a fair price. Third, if users shop for presents, then we can gain some information about their family background, e.g. people shopping for toys will most likely have a child themselves or among their closer relatives. Our findings indicate that this is indeed the case. Table 1 summarizes the data set in its basic parameters.

3. User activity and user networks

The activity of the users is measured via the probability mass distributions of the number of articles sold $p(s)$, bought (auctions won) $p(w)$, and bid on $p(a)$. Though it is possible to bid multiply in a single auction, we neglect this fact and use ‘bid’ and ‘take part in an auction’ synonymously. Similar to previous studies [2], we find fat tailed distributions of the user activity. Due to the short time span observed and a constant growth of the market, one cannot regard these distributions as representing a steady state. Nevertheless, some insight can be obtained. We compared maximum likelihood fits of the data to log-normal distributions of the form $p(x) \propto (x-\theta)^{-1} \exp[-(1/2)(\ln((x-\theta)/m)/\sigma)^2]$ and power laws of the form $p(x) \propto (x + \Delta x)^{-\kappa}$ and find that both kinds of distribution characterize the data almost equally well with a slight advantage for the power law especially for the very rare events in the tail of the distribution as can also be observed in the cumulative plots. In [2], only power laws were considered. For the number of bidders b taking part in an auction, the ‘attractiveness of an article’, we consider an exponential distribution $q(b) \propto \alpha^b$. A possible alternative distribution, the Binomial or Poisson distribution can be ruled out, as the empirical distribution is monotonously decreasing and these distributions would require a maximum at the average value. Alternatively, an almost perfect fit can be achieved when assuming $q(b) \propto \exp(-b^{1.117}/3.7)$. However, we prefer the simple model with only one free parameter $\alpha = 1 - 1/\langle b \rangle$ and attribute the observed deviation at high numbers of bidders per article to a saturation effect. We believe that if an article has attracted a critical number of bidders, potential additional bidders are more reluctant to join because of the already strong competition and hence there are up to ten times less articles with 15 or more bidders than expected from the simple model. Recall that only one in 1000 auctions attracts more than 15 bidders. Naturally, our hypothesis would have to be confirmed or rejected by future research. Figure 2 shows a graphical representation of these distributions and table 2 summarizes the parameters obtained by maximum likelihood fitting [19, 20].

Clustering of sparse data via network communities

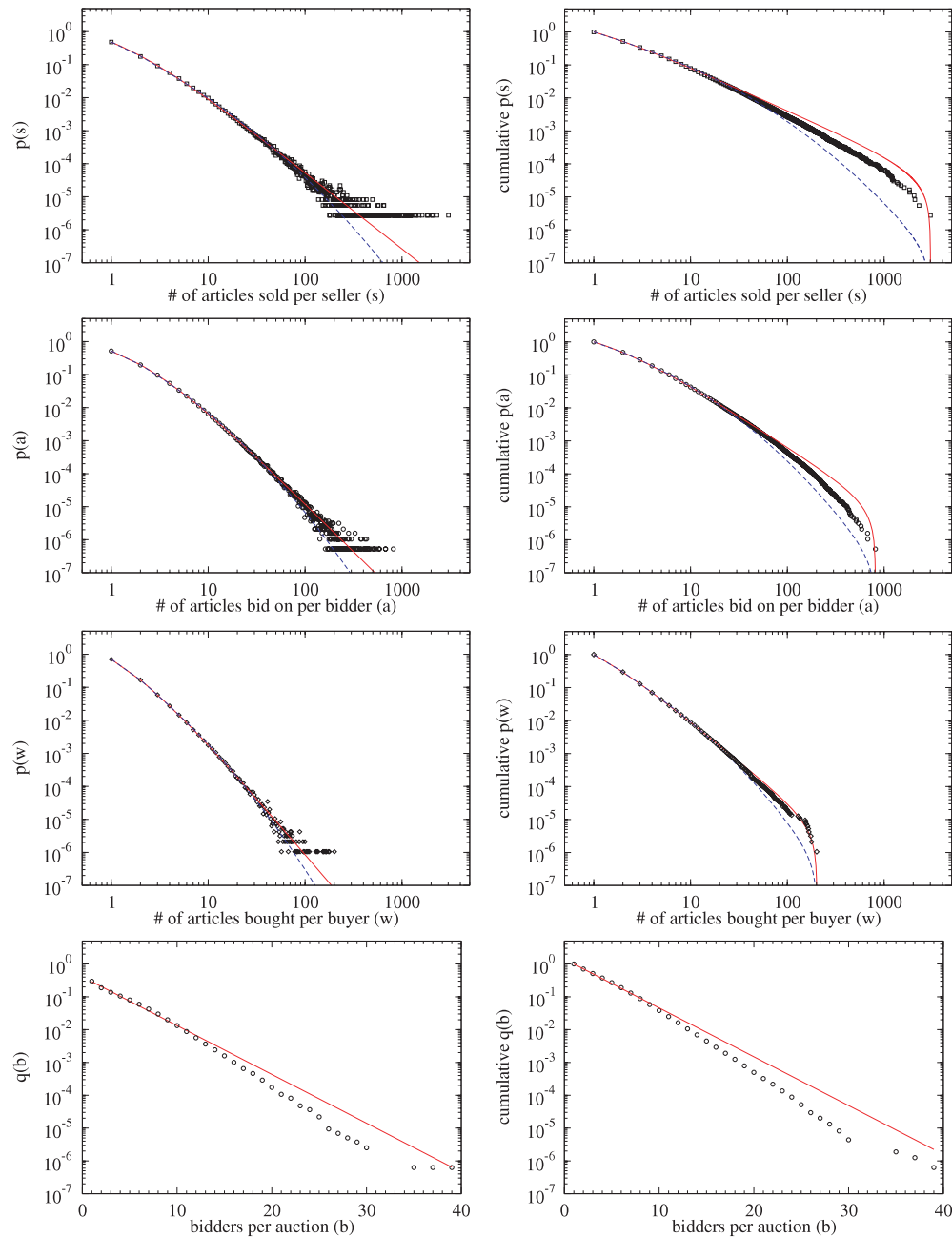


Figure 2. User activity during the pre-Christmas season 2004. From top left to bottom right: probability mass function (left) and cumulative probability distribution (right) of the number of articles sold $p(s)$, different auctions participated in $p(a)$ and number of articles bought $p(w)$ as well as number of bidders participating in an auction $q(b)$. For the first three distributions, the red solid lines correspond to maximum likelihood fits with power laws $p(x) \propto (x + \Delta x)^{-\kappa}$ while the blue dashed lines correspond to fits with a log normal distribution of the form $p(x) \propto (x - \theta)^{-1} \exp[-(1/2)(\ln((x - \theta)/m)/\sigma)^2]$. The fits for the cumulative distribution take the finite size of the sample into account. For the distribution of the desirability of an article $q(b)$, the solid red line represents a maximum likelihood exponential fit of the form $q(b) \propto \alpha^b$. Parameter estimates and log likelihood scores can be found in table 2.

Table 2. Estimated parameters of activity distributions of observed users in auctions. For the distribution of the number of articles sold per seller $p(s)$, the number of auction taken part in by a bidder $p(a)$ and the number of articles bought by a buyer $p(w)$, we show the maximum likelihood estimates of the parameters for both a log-normal distribution $p(x) \propto (x - \theta)^{-1} \exp[-(1/2)(\ln((x - \theta)/m)/\sigma)^2]$ as well as for a shifted power law of the form $p(x) \propto (x + \Delta x)^{-\kappa}$. Further, we indicate the likelihood scores of these distributions for the data. Note that all distributions are better described as power laws, but the difference to the log-normal is very small. For the distribution of the number bidders taking part in an auction, we assume an exponential form $q(x) \propto \alpha^x$ and show the ML estimate of the parameter α . Additionally, the averages of all quantities are given.

$p(x)$	$\langle x \rangle$	θ	m	σ	L_{LN}	Δx	κ	L_{Pow}	α
$p(s)$	4.3	0.93	1.28	1.55	-1.9427	0.96	2.31	-1.9419	
$p(a)$	2.9	0.82	1.08	1.32	-1.6722	1.55	2.90	-1.6719	
$p(w)$	1.7	0.90	0.60	1.21	-1.0415	0.90	3.43	-1.0414	
$q(b)$	3.4								0.71

The distribution of the number of articles sold per seller falls off slowest, followed by the number of articles bid on and the number of articles bought. Here, we see the professionalization on the seller side of the market. There are ‘power-sellers’ making a living from selling via eBay, but there are hardly any ‘power-buyers’ professionally buying on eBay. This shows that eBay is more of a selling platform than an actual trading site, where selling and buying activities would be more balanced. If we assume that the tail of the distribution of the number of articles sold per seller is representative for the ‘firm size’ of these users and compare these to the long term statistics of firm sizes in the US given by Axtell [21], we can confirm the power law tail of the distribution, but not the exponent of $\kappa = 2$. Instead, we find $\kappa = 2.31$ and thus the observed distribution falls off faster. We can only speculate on the reasons for this and further study is needed here to compare new and old economy. In an earlier study, Yang *et al* had reported an exponent of $\kappa = 3.5$ for the distribution of the number of auctions a bidder takes part in from a data set obtained in 2001 [2] and we found $\kappa = 2.9$ in our data. If this discrepancy is the result of a trend and not due to the differences in the observed countries and sizes of the data set, and this trend holds also for the distribution of the seller’s activity, then one may be able to observe a convergence towards the exponent of $\kappa = 2$ known from the old economy. Further study is needed regarding this hypothesis.

The fat tails of the distribution are striking given the short time span observed. Consider the most active bidder taking part in over 800 auctions! This user seems to follow a gambling strategy bidding only minimal amounts as he/she wins only a few of these auctions. The most successful buyer who won 201 auctions on the other hand took part in only 208 auctions. This hints at a diversity of strategies employed by users of the online auction site. Curiously, the article most desired and attracting 39 different bidders was a ride in a red Coca-Cola truck. The fat tails of the distribution also show that there is no ‘typical’ user activity, rather, one observes activities at all scales.

Table 3. Summary of basic parameters for the bidder network with two bidders linked, if they have competed in an auction. Shown are the actual data, the parameters for a random null model (RNM) obtained by reshuffling the bidders in different auctions and the reduced version of the network used for cluster analysis containing only those bidders having taken part in more than one auction.

	Data	RNM	Reduced
Number of nodes	1.8×10^6	1.8×10^6	0.9×10^6
Number of links	11.6×10^6	12.6×10^6	7.4×10^6
Average degree	12.9	13.9	16.4
Assortativity	0.02	$-(4 \pm 3) \times 10^{-4}$	0.03

From the original data a number of market networks can be constructed, such as the network of users connected by actual transactions, or the network of sellers that are connected if they have sold to (or received bids from) the same user. Then, the links in the network would represent competition or a possibility for cooperation, depending on the portfolio of articles offered by these sellers.

Here, we focus only on the bidder network based on single articles. Two bidders are linked if they have competed in an auction. Since all users who bid in a single auction are connected, this network results from overlaying fully connected cliques of bidders that result from each auction. Such graphs are also known as affiliation networks [22]–[24].

Prior to a cluster analysis in this bidder network, we study its general statistical properties looking for indications of cluster structure [25]. We compare the results to a randomized null model (RNM) obtained from reshuffling the original data, i.e. keeping the attractiveness of each auction and the activity of each bidder constant, but randomizing which bidders take part in which auction. Figure 3 shows a comparison between the empirical data and the RNM in terms of the cumulative degree distribution, the cumulative distribution of the link weights as well as the clustering coefficient $c(k)$ as a function of the degree k .

The shapes of the cumulative degree distributions agree quite well between the RNM and the original data. However, for higher values of k , the distribution of the empirical data lies below that of the RNM. Since meeting the same competitor twice in different auctions does not lead to an increase in the number of neighbours in the network, this shows that competitors meet more often in the real world than expected from the random null model. A theoretical expectation for the average number of neighbours in the bidder network is be given by $\langle k \rangle = 2(\langle b \rangle - 1)\langle a \rangle = 14$ where $\langle b \rangle$ is the average number of bidders per auction and $\langle a \rangle$ is the average number of auctions taken part in by a bidder. This estimate is in excellent agreement with the result from the RNM ($\langle k \rangle = 13.9$), but larger than in the actual data ($\langle k \rangle = 12.9$) confirming our expectation. See table 3 for a summary of the basic parameters of the empirical data and the RNM.

Comparing the cumulative distribution of the link weights, i.e. the number of times two bidders have met in different auctions, we find a much more prominent difference between the data and the RNM. Figure 3 shows that the weights of the links in the bidder network are distributed with a power law tail. Approximately 6% of all links correspond to pairs of bidders who have met more than once. If there would be no common interest

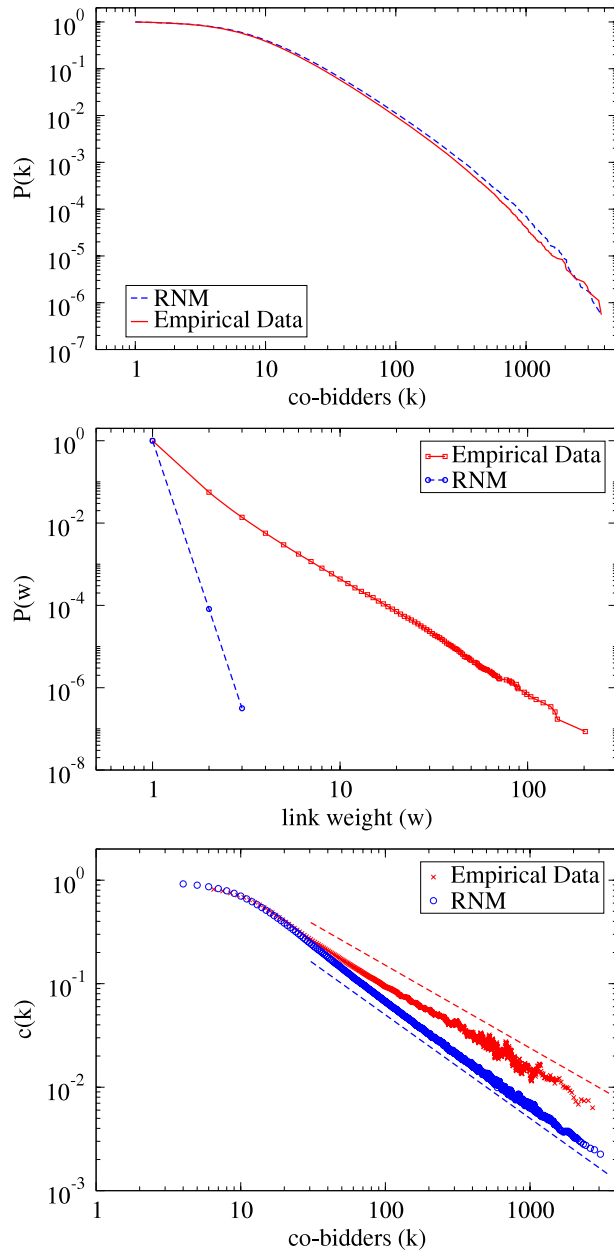


Figure 3. Comparison of the bidder network with a random null model (RNM). Top: cumulative degree distribution $P(k)$. The degree distribution found empirically (solid red line) lies below that of the random null model (dashed blue line). Middle: cumulative distribution of the link weights in the bidder network. The distribution for the empirical data (solid red line) lies above that of the random null model (dashed blue line). Bottom: distribution of the clustering coefficient $c(k)$ as a function of the degree k of the nodes. The clustering is higher for the empirical data (red crosses) than for the RNM (blue circles). The two dashed lines indicate two power laws $\propto k^{-\kappa}$ with exponents $\kappa = 1$ and 0.8 , respectively, for comparison.

among bidders, practically all links would have weight 1 as is indeed the case for the RNM.

Additionally to the distribution of degrees and link weights, we compare the distribution of the clustering coefficient as a function of the degree of a node. The clustering coefficient $c(k)$ denotes the average link density among the neighbours of a node of degree k . Due to the construction process of the network as an affiliation network, we expect that for large numbers of neighbours k the clustering coefficient $c(k)$ scales as k^{-1} in case of random assignment of bidders to auctions [23]. Figure 3 shows that this is indeed the case for the RNM, but the actual data deviate strongly for bidders with a large number of neighbours and shows higher clustering. This effect can arise from two processes: either bidders with whom one competes in two different auctions also meet independently in a third auction, or that there is an increased probability that one will compete again with a bidder one has already met once in an auction. Both explanations support our assumption of the presence of clusters of users with common interest.

With these comparisons, we have shown that the bidder network is far from randomly constructed and we will proceed by studying the cluster structure for which we have found indirect evidence already.

4. Market segmentation

4.1. Network clustering

The analysis of the user interests in the eBay market is based on the bidder network as constructed in the previous section. The links in this network represent articles the connected bidders (nodes) have a common interest in. Since we are eventually interested in the overlap of user interests, we reduced the full network to only those bidders having taken part in at least two auctions and considered only auctions with a final price below 1000 Euro, a price range in which we expect users to consider bidding on and buying several items. Though this reduction takes out the least active bidders from the data set, we believe that in order to sensibly classify a user according to its bidding interests during a certain time span, there should be a minimum number of observations of bidding activity. See table 3 for the basic parameters of this reduced network.

If we now find groups of users (clusters or communities [26]–[28]) with a high density of links among themselves and a low density of links to the rest of the network, the total set of links within such a group of users can be interpreted as a unifying common interest of this group. We assign a community or cluster index to each bidder as to maximize a well established quality function known as network modularity Q defined by Girvan and Newman (GN) [29]. The definition of Q can also be written as [30]:

$$Q = \frac{1}{M} \sum_s^q \underbrace{(m_{ss} - \gamma[m_{ss}])}_{c_{ss}} = -\frac{1}{M} \sum_{s < r}^q \underbrace{(m_{rs} - \gamma[m_{rs}])}_{a_{rs}}. \quad (1)$$

Here, the first sum runs over all group indices s , while the second over all pairs of different group indices $s > r$, m_{ss} is the number of internal links in group s , i.e. connecting bidders with the same group index s . Accordingly, $[m_{ss}]$ is an expectation value for this quantity assuming a random assignment of bidders into groups of the same size and degree distribution and is given by $[m_{ss}] = K_s^2/4M$. By K_s we denote the total number of links

emanating from members of group s , while M is the total number of links in the network. Equivalently, m_{rs} is the number of links between members of group r and s and $[m_{rs}]$ is the corresponding expectation value given by $[m_{rs}] = K_s K_r / 2M$ [30]. Q is maximal, when the sum of cohesions c_{ss} , defined as the difference between the actual and expected number of within group links, is maximal. Equivalently, Q is maximal when there are many links less between different groups than expected for a random assignment of nodes into communities, i.e. the sum of adhesions a_{rs} is minimal. Note that any assignment of bidders into groups which maximizes Q will be characterized by both, maximum cohesion of groups, and minimal adhesion between groups. If Q is maximal, every node is classified in that group to which it has the largest adhesion, otherwise it could be moved to a different group to increase Q . Additionally to the original definition of Q by GN, we have introduced a parameter γ which allows to adjust the relative influence of actually present and expected links in the definition (setting $\gamma = 1$ recovers the original definition of GN). At any (local) maximum of Q , the cohesion c_{ss} of a group is never negative and the adhesion between different groups a_{rs} is never positive. Hence, values of γ smaller or greater than one lead to larger or smaller communities, respectively. Comparing classifications obtained at different values of γ allows the detection of hierarchy and overlap in the cluster structure. See [31, 30] for examples and further details of this variation.

The technical details of how the bidders can be assigned into groups such that Q is maximized using simulated annealing are given in [30, 27, 32, 33]. Note that the optimization routine exploits the sparsity of the data as it only operates on the links of the network. We allowed for maximally $q = 500$ different groups of bidders in our analysis which gives a sufficient level of detail.

The left part of figure 4 compares the results obtained with $\gamma = 0.5$ and 1. Shown are the adjacency matrices A_{ij} of the largest connected component of the bidder network. A black pixel at position (i, j) and (j, i) is shown on an $889\,828 \times 889\,828$ square if bidders i and j have competed in an auction and hence $A_{ij} = 1$, otherwise the pixel is left white corresponding to $A_{ij} = 0$. The rows and columns are ordered such that bidders who are classified as being in the same group are next to each other. The internal order of bidders within groups is random. The order of the groups was chosen to optimally show the correspondence between the ordering resulting from the $\gamma = 0.5$ and the $\gamma = 1$ ordering. In this representation, link densities correspond to pixel densities and thus to grey levels in figure 4. Information about the exact size and link density contrast of the clusters is given in table A.1. Note the high contrast between internal and external link density.

At the top of figure 4, we show the adjacency matrix ordered according to an optimal assignment of bidders into groups with $\gamma = 0.5$. Clearly, a small number of major clusters of bidders and a large number of smaller clusters are identified, strongly connected internally and well separated from one another. The largest 8 clusters are marked with letters A–H. Of all bidders in the network, 85% are classified in these eight clusters. At the bottom of figure 4, we show the same adjacency matrix, but now rows and columns are ordered according to an optimal assignment of bidders into groups with $\gamma = 1$. As expected, we find a larger number of smaller, denser clusters, the largest of which are numbered 1–13. In order to analyse whether the network has a hierarchical or overlapping cluster structure, we define a consensus ordering of the bidders from the $\gamma = 0.5$ and 1 ordering by reshuffling the internal order of the $\gamma = 0.5$ clusters according to the $\gamma = 1$ clustering. Remember the orderings for the two values of γ were obtained independently.

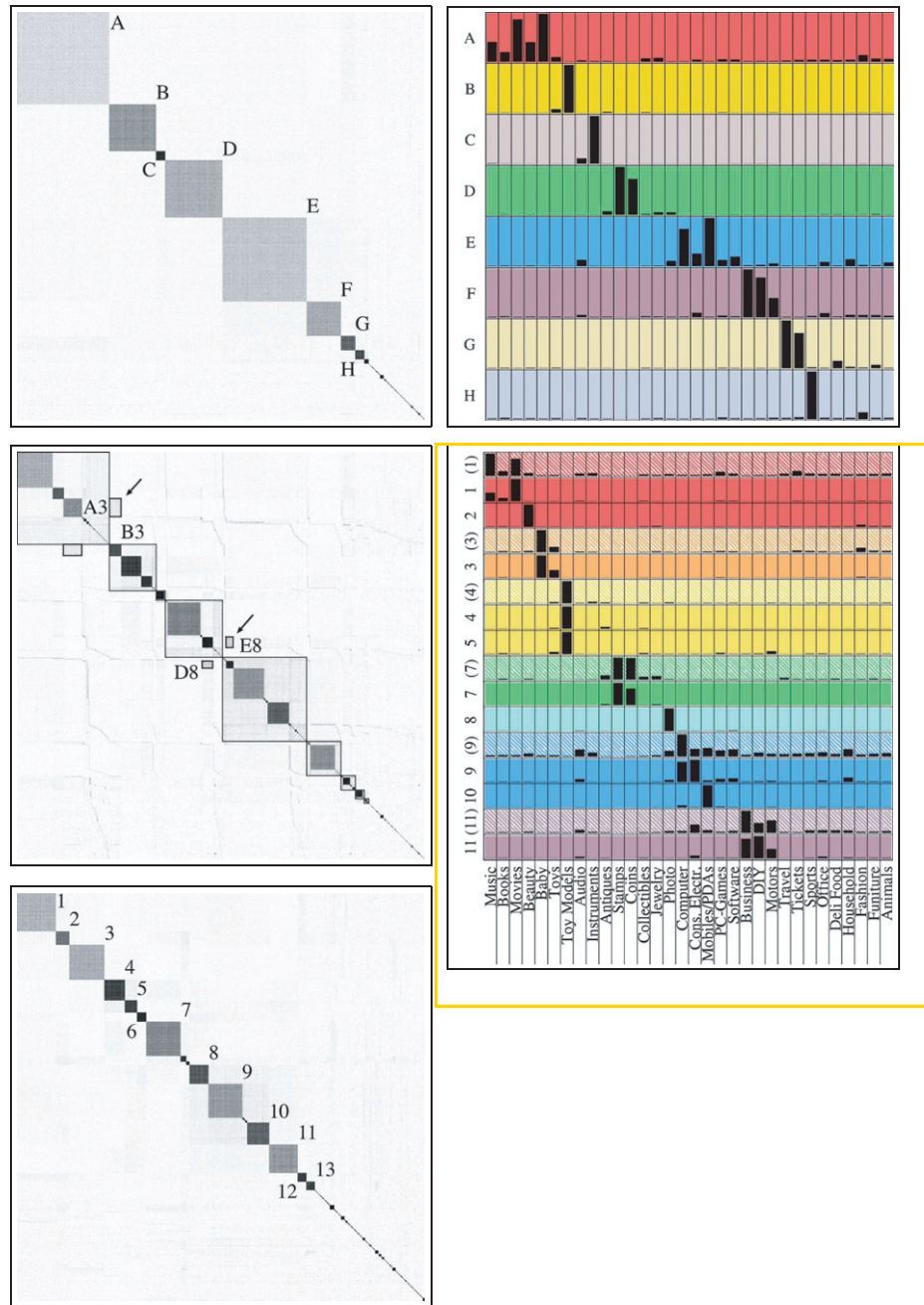


Figure 4. Left: $N \times N$ adjacency matrix of the bidder network in three different orderings. A pixel in row i , column j corresponds to an auction in which bidder i and j have competed. Shown are $N = 889\,828$ bidders (nodes) and $M = 7373\,008$ pairwise competitions (links). Grey levels correspond directly to link density in this network and hence to the probability of competing in an auction. Top: $\gamma = 0.5$ ordering, bottom: $\gamma = 1$ ordering and middle: consensus ordering of top and bottom. Right: odds ratios of bidding in one of the 32 main eBay product categories for classified users (normalized to maximum value for each cluster). Top: from $\gamma = 0.5$ classification, bottom: from $\gamma = 1$ classification. Spectra with a dashed background (cluster id in parenthesis) show customer purchases 6–9 months after original classification. See text for details.

If the network possesses a hierarchical structure in the sense that the clusters obtained at higher values of γ lie completely within those obtained at lower values of γ , then the consensus ordering would not differ from the ordering at $\gamma = 1$. If, however, clusters at lower values of γ overlap and this overlap forms its proper cluster at higher values of γ , the network is not entirely hierarchical. These aspects will become immediately clear by looking at the middle part of figure 4. For clarity, we have marked the borders of the $\gamma = 0.5$ clustering. Clusters 1 and 2 fall entirely within cluster A giving an example of a cluster hierarchy. Cluster 3, however, is split by the consensus ordering into one part A3 belonging to A, and B3 belonging to B (see arrows in figure). It is now clear that clusters A and B actually have some overlap which was not visible in the $\gamma = 0.5$ ordering due to the random order of all bidders with the same cluster index. This overlap is concentrated in cluster 3, parts of which belong stronger to either A or B. Clusters 4 and 5 then fall again completely within cluster B. Clusters 6 and C are practically identical. Cluster D has a number of sub-clusters, the largest of which is 7 and overlaps with cluster E through cluster 8 as before (see arrows again). Group E has two more sub-groups 9 and 10 while clusters 11, 12 and 13 fall entirely within clusters F, G and H, respectively. More details about hierarchical and overlapping cluster structures including some toy examples can be found in [30].

4.2. Cluster validation, interpretation and time development

To validate the statistical significance and to rule out the possibility the observed cluster structure is merely a product of the clustering algorithm or the particular method of constructing the network from overlapping cliques of bidders, we need to compare the results to those obtained for appropriate random null models [34]. Maximizing Q also for the RNM version of the bidder network, again taking into account those bidders who took part in at least two auctions, we find a value of $Q = 0.28$ at $\gamma = 1$ which is significantly less than the value of $Q = 0.64$ for the empirical data. Furthermore, the RNM shows all equal sized clusters, while the real network clearly possesses major and minor clusters. A random graph with the same number of nodes and links, i.e. disregarding the scale free degree distribution and the affiliation network structure of the graph, would yield only $Q = 0.23$ [30]. Though this analysis does not give a quantitative measure of the statistical significance of the cluster analysis it certainly indicates a strong deviation from the RNM. The analysis of the temporal development of the user interests below will provide additional validation.

Until now we have only found groups of bidders that have an increased probability to meet other members of their groups in the auctions they take part in. The eBay product categories are now used in order to find an *interpretation* for the common interests that lead to the emergence of the cluster structure of the bidder network. Since cluster sizes vary and the number of articles in the individual categories is very diverse, we calculate the odds ratios (OR) for bidding in one of the 32 main categories. This odds ratio is defined as

$$\text{OR}_{Cs} = \frac{P(\text{bidding in } C | \text{member of cluster } s)}{P(\text{bidding in } C | \text{not member of cluster } s)}, \quad (2)$$

i.e. the ratio of the odds of bidding in category C , given a bidder is member of group s versus the odds of bidding in category C given the bidder is member of any group $r \neq s$.

The right-hand side of figure 4 shows a graphical representation of the odds ratios for clusters A–H and most of the clusters 1–13. All spectra are normalized to their maximum value. The exact non-normalized numerical values can be found in table A.2. Clusters from the $\gamma = 1$ assignment are more specific with less entries in the category spectrum and larger ORs.

Cluster A unites bidders interested in articles listed in the baby, beauty, fashion, books, movies and music category. Cluster 1 then represents a more specifically content oriented user group mainly interested in books, movies and music. As we have seen, cluster 1 is an almost complete sub-cluster of A. Cluster 2 is also a complete sub-cluster of A and encompasses bidders mainly interested in cosmetics and fashion.

Cluster B contains two sub-clusters 4 and 5, both annotated in the toy model category. Closer inspection, however, reveals that cluster 4 is mainly characterized by its interest in model railways while the bidders in cluster 5 have a passion for model cars, radio controlled models, slot cars and the like. Note the advantage of clustering based on single articles. The clusters we find with one simple unbiased method combine top level categories as in the case of cluster 1 or can only be described by resorting to sub-categories as in the case of clusters 4 and 5. From the left part of figure 4, we had observed that cluster 3 is responsible for a large part of the overlap between clusters A and B. We see that users in this group 3 have their main interests in the baby and toy category. The overlap of cluster A and B is hence mediated via the toy category. Members of cluster A and B mainly meet in toy auctions. The interpretation of the other clusters is then equally straightforward.

Bidders in clusters C and the practically identical cluster 6 take interest in audio equipment and instruments. Cluster D represents bidders with an inclination to collecting, their bids being placed in the antiques, jewelry, stamps and coins category (cluster 7). The bidders in cluster E are mainly shopping for technological gadgets, computers, consumer electronics, software, mobile phones, PDAs etc (clusters 9 and 10). Their overlapping interest with bidders from cluster D is in items from the photo category (cluster 8). In groups F and 11, we find predominantly practically oriented users who place their bids mainly in the categories of automotive spare parts, business and industry (where a lot of tools and machinery are auctioned) and do-it-yourself. Finally, in groups G and 12 we find event oriented customers with strong bidding activity in the tickets and travel category and in group H and 13, we find people bidding on sports equipment.

Let us now focus on the time development of the user interests. The data for our analysis were collected during a relatively short time span (25 days) and we base our results on an extremely sparse data set. Remember that every bidder in the network took part in only 3 auctions on average. Is it really possible to predict meaningful patterns of consumer interest from such sparse data? One could further argue that the few most active bidders account for a large portion of the bids, thus holding the network together and ‘defining’ the clusters of interest, because they also contribute a large number of links. In order to address this question, we revisited the data set in the beginning of September 2005, more than nine months after our original study. From the six largest clusters of the $\gamma = 1$ ordering, we uniformly and randomly sampled 10 000 users each. Note that this removes possible bias towards very active users, they are now represented in the data according to their proportion in the population. Then we looked at the trading history of these users as far back as eBay permits—90 days. For these 60 000 users, we determined the product categories of the articles they had bought between June and September.

Again, we calculated the odds ratios, this time of *buying*, i.e. winning an auction, from a particular category and with the new sample of users as basic population. The results are shown on the right-hand side of figure 4 with a dashed background and the cluster id from which the users were sampled in parenthesis. The stability of the interest profiles is quite remarkable. The main interests have remained unchanged as compared to the initial study though in some cases the spectrum has become more diverse. For instance the content oriented bidders of cluster 1 now also show increased buying activity in the PC-games and tickets category. At the same time the main interest has shifted from movies to music. The largest number of product categories with increased odds of bidding in this category is found for cluster 9, the members of which are the most technology affine users anyway and which would be expected to satisfy a very broad range of consumer needs from online vendors. The members of cluster 7 (the collectors) and cluster 4 (the toy model builders) are much more conservative and almost do not change their profile at all. Without second hand data about the age structure of the bidders classified, we can only speculate that these clusters are formed by older customers who tend to stick to particular categories.

5. Conclusion

Employing a recently developed network clustering technique, we have presented a detailed study of the user behaviour on the online auction site www.ebay.de during the pre-Christmas season of 2004. Fat tailed distributions of user activity in terms of the number of articles sold, bought, and bid on were found. The attractiveness of articles, measured in terms of the number of bidders participating in an auction, shows an exponentially decaying distribution. Focusing on the bidding behaviour, we constructed a network of bidders from their competition for single articles. Nodes in the network correspond to bidders and links to the fact that these bidders have expressed a common interest in at least one article. Studying the general statistical properties and comparing to appropriate random models, we find clear indications for a non-trivial cluster structure. This cluster structure, its hierarchy and overlap were studied using a community detection algorithm. Our analysis did not need the definition of any kind of measure of similarity between articles or product categories. Rather, we solely used the taxonomic information about articles provided by eBay to interpret our results. We can classify 85% of the users into only a small number of well separated, large clusters, all of which have a distinct profile of only a few main interests as revealed by annotating the articles in the taxonomy of product categories. Some of the clusters show sub-clusters or overlap with other clusters. The interest profiles we identified are remarkably stable. Sampling randomly from the clusters and checking, what these users bought during a three month period in the summer 2005, we found that the profiles of articles bought were almost identical to those from the classification six months earlier.

This is striking because virtually everything is offered on eBay and one would expect users to satisfy a much broader range of shopping interests. However, it appears that the major clusters mainly correspond to people's favourite spare time activities. We believe the apparent stability of users' buying and bidding behaviour reflects the permanence of their interests which is also stabilized by their social environment and activities. The clear signature in the market data may stem from the fact that users tend to buy online only articles where they have some experience and expertise. Users seem hesitant to bid on

articles from categories in which they have not previously bid. This may be due to the fact that inexperienced users cannot judge what is a fair price for an article in an auction and they have difficulty assessing to what extent the article offered really suits their needs. At the same time, user interests are reinforced by online recommender systems [35, 36], which suggest similar articles to those already bought by the user. Here lies enormous growth potential for the auction site by extending the way in which people use the site, i.e. to facilitate buying goods the user has no experience with and the introduction of ‘eBay Express’ or the ‘Buy it now’ option are developments in this direction.

On the other hand, the profiles found and their temporal stability corroborate the hypothesis that the presence of latent interest profiles in the society *per se* leads to the emergence of user groups with common interest. Transparent markets such as online auction sites in which users act independently and anonymously are perfect starting points for tapping into this collective behaviour and methods from physics can provide the tools leading to its understanding.

Appendix. Cluster parameters

Table A.1. Summary of basic parameters for the major communities found in the bidder network (annotated as in figure 3). N denotes the number of bidders in the cluster, $\langle k_{\text{in}} \rangle$ and $\langle k_{\text{out}} \rangle$ the average numbers of neighbours within the cluster and in the rest of the network, respectively. By p_{in} and p_{out} we denote the internal and external link density, respectively. The average link density in the network is $\langle p \rangle = 1.9 \times 10^{-5}$.

Cluster	N	$\langle k_{\text{in}} \rangle$	$\langle k_{\text{out}} \rangle$	p_{in}	p_{out}
A	200 630	10.2	3.4	5.1×10^{-5}	5.0×10^{-6}
1	846 99	10.3	4.0	1.2×10^{-4}	5.0×10^{-6}
2	293 23	9.0	5.2	3.1×10^{-4}	6.0×10^{-6}
3	761 82	10.1	4.1	1.3×10^{-4}	5.0×10^{-6}
B	102 188	18.6	3.9	1.8×10^{-4}	5.0×10^{-6}
4	448 30	24.6	4.2	5.5×10^{-4}	5.0×10^{-6}
5	263 25	14.2	5.2	5.4×10^{-4}	6.0×10^{-6}
C	199 15	14.1	4.3	7.1×10^{-4}	5.0×10^{-6}
6	200 20	14.5	4.3	7.3×10^{-4}	5.0×10^{-6}
D	124 702	16.5	3.8	1.3×10^{-4}	5.0×10^{-6}
7	749 13	17.2	4.1	2.3×10^{-4}	5.0×10^{-6}
8	413 59	16.8	5.9	4.1×10^{-4}	7.0×10^{-6}
E	183 313	15.4	4.2	8.4×10^{-5}	6.0×10^{-6}
9	737 22	13.4	6.5	1.8×10^{-4}	8.0×10^{-6}
10	479 37	17.5	5.9	3.7×10^{-4}	7.0×10^{-6}
F	746 57	10.5	4.9	1.4×10^{-4}	6.0×10^{-6}
11	621 15	11.1	5.0	1.8×10^{-4}	6.0×10^{-6}
G	313 37	11.0	6.0	3.5×10^{-4}	7.0×10^{-6}
12	188 35	11.8	6.1	6.3×10^{-4}	7.0×10^{-6}
H	196 20	10.0	4.4	5.1×10^{-4}	5.0×10^{-6}
13	182 86	9.9	4.4	5.4×10^{-4}	5.0×10^{-6}

Table A.2. Odds ratios for *bidding* in one of the 32 main categories during the pre-Christmas season 2004. Shown are only values above 1 signifying an increased interest in articles from this category. Values larger or equal 5 are set in bold font. Note how the clusters from the $\gamma = 1$ clustering (1–13) are more specific than those from the $\gamma = 0.5$ clustering (A–H), e.g. there are less categories with an OR larger 1 and those that are deviate stronger from 1. Also note how the overlap between cluster A and B is mediated via the toy category and the overlap between cluster D and E via the photo category. For the six largest clusters of the $\gamma = 1$ clustering, we show the OR of *buying* from the corresponding category during the summer 2005 as explained in the text.

Category	A	1	2	3	B	4	5	C	D	7	8	E	9	10	F	11	G	H
Music	5.6	8.6 (12.0)						1.5										
Books	2.7	3.4 (2.5)		(1.1)														
Movies	12.1	22.1 (9.2)																
Beauty	5.6	(1.5)	29.3	(1.3)									(1.7)					
Baby	13.7			41.0 (20.0)														
Toys	1.3			14.2 (4.9)	5.0	(1.6)	1.1											
Toy models					70.2	21.9 (28.0)	10.2											
Audio		(1.3)						16.2			1.2	3.4	2.1 (3.9)		1.0	1.4 (2.3)		
Instruments		(1.5)				(1.9)		144.3					(2.2)					
Antiques					1.2	1.8 (1.1)			6.6	6.3 (3.7)								
Stamps									88.6	118.0 (18.0)								
Coins									66.7	88.3 (18.0)								
Collectibles									1.6	1.7 (2.6)								
Jewelry	1.1		1.3						4.5	5.3 (3.1)								

Table A.2. (Continued.)

Category	A	1	2	3	B	4	5	C	D	7	8	E	9	10	F	11	G	H
Photo						(1.2)			4.4		47.9	2.8	1.5 (3.2)			(1.4)		
Computer												19.6	14.7 (12.0)	3.1		(1.4)		
Cons. electr.												6.7	16.1 (4.3)		2.0	2.5 (5.9)		
Mobile												25.2	1.3 (4.7)	33.0		(2.1)		
Games		(2.2)		(1.1)								3.3	2.3 (3.4)	1.4				
Software		(1.2)									1.5	5.1	2.7 (4.0)			(1.5)		
Business													(1.1)		21.5	16.9 (15.0)		
DIY													(2.2)		17.9	19.1 (7.0)		
Motors							1.4					1.5	(1.6)		8.6	8.2 (8.9)		
Travel		(1.2)								(1.7)			(1.4)				58.2	
Tickets		(2.7)		(1.7)									(1.7)				42.9	
Sports		(1.3)		(1.3)									(2.0)			(1.9)	1.4	16.0
Office		(1.1)										2.3	1.1 (2.4)		1.9	1.6 (2.0)		
Deli food		(1.2)											(1.5)			(2.0)	9.0	
Household												3.7	3.3 (4.1)		1.1	(1.8)	1.8	
Fashion	1.9		2.7	1.7 (4.1)									(1.2)				1.1	2.3
Furniture		(1.0)	1.1	(1.6)					1.3				(1.6)				4.2	
Animals		(1.2)	1.0	(1.3)								2.0	(1.9)			(1.4)		

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